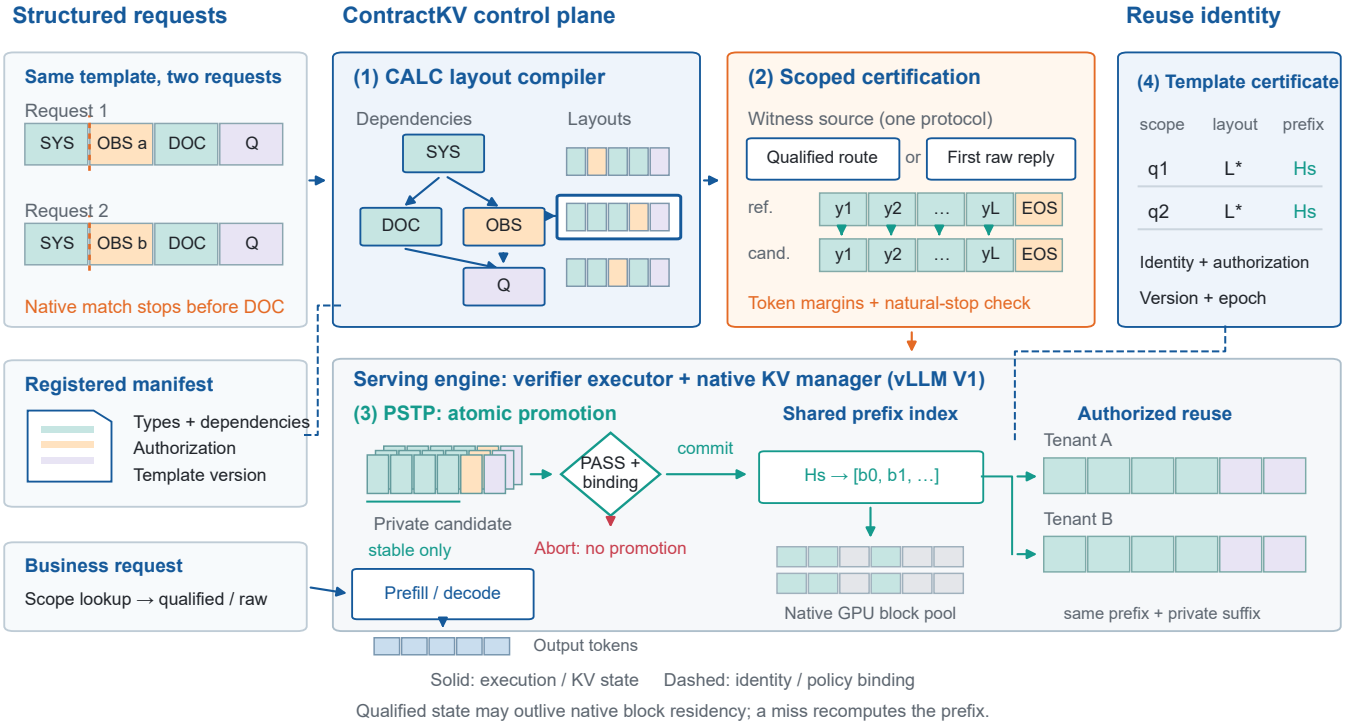

Structured requests
ContractKV control plane
Reuse identity
Same template, two requests
Request 1
SYS
OBS a
DOC
Q
Request 2
SYS
OBS b
DOC
Q
Native match stops before DOC
(1) CALC layout compiler
Dependencies
Layouts
SYS
DOC
OBS
Q
(2) Scoped certification
Witness source (one protocol)
Qualified route
or
First raw reply
ref.
cand.
y1
y2
…
yL
EOS
Token margins + natural-stop check
(4) Template certificate
scope
layout
prefix
q1
L*
Hs
q2
L*
Hs
Identity + authorization
Version + epoch
Registered manifest
Types + dependencies
Authorization
Template version
Serving engine: verifier executor + native KV manager (vLLM V1)
(3) PSTP: atomic promotion
Private candidate
stable only
PASS + binding
commit
Abort: no promotion
Shared prefix index
Hs → [b0, b1, …]
Native GPU block pool
Authorized reuse
Tenant A
Tenant B
same prefix + private suffix
Business request
Scope lookup → qualified / raw
Prefill / decode
Output tokens
Solid: execution / KV state
Dashed: identity / policy binding
Qualified state may outlive native block residency; a miss recomputes the prefix.